Office of the Director General of Police
Commandant General, Home Guards & Director of Civil Defence and Director General Karnataka State Fire & Emergency Services
No. 1, Annaswamy Mudaliar Road
Banglore - 560 042

Phone : 25570733
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KARNATAKA STATE FIRE & EMERGENCY SERVICES

NO OBJECTION CERTIFICATE

No. KSFES/GBC(1)/580

Docket No. KSFES/NOC/530/2024 Dated : 07/03/2025 10:36

To,

The Chief Commissioner,
BBMP,
Corporation circle,
Hudson Circle,
Bangalore - 560 002.

Sir

Sub : Issue of No Objection Certificate for the construction of Residential building at AT SY NO. 55/1, 55/2, 55/3, 55/4, 56/2, 56/3, 62/1, 62/4, 62/5, 63/2, 63/3, 63/4, 63/5, 63/6 GUNJUR VILLAGE, VARTHUR HOBLI, Bangalore East, BANGALORE - 560087

Ref:

1. Letter dated 25/12/2024 of the Authorized Signatory, M/S. BRICKS & MILESTONES PROJECTS LLP 1140, VGR ESSOR, 3RD FLOOR, 6TH MAIN, 17TH CROSS, 7TH SECTOR, HSR LAYOUT, Bangalore South, BANGALORE - 560102

With reference to the letter of the M/S. BRICKS & MILESTONES PROJECTS LLP 1140, VGR ESSOR, 3RD FLOOR, 6TH MAIN, 17TH CROSS, 7TH SECTOR, HSR LAYOUT, Bangalore South, BANGALORE - 560102 cited above, the Regional Fire Officer, CFO - Bangalore East Zone of this department has inspected the site of proposed Residential buildings at AT SY NO. 55/1, 55/2, 55/3, 55/4, 56/2, 56/3, 62/1, 62/4, 62/5, 63/2, 63/3, 63/4, 63/5, 63/6 GUNJUR VILLAGE, VARTHUR HOBLI, Bangalore East, BANGALORE - 560087 on 08/01/2025 02:00 with reference to the drawings furnished by the builder and the details are as follows:

	Part-A: General Building requirements.	
1	Address of Applicant	**Authorised Signatory,** **M/s. Bricks & Milestones – Projects LLP,** **VGS ESSOR,** No. 1140, 4 th floor, 17 th Cross, 7 th Sector, HSR Layout,

		Bangalore – 5600102.
2	Address of the Premises	Sy. Nos. 55/1, 55/2, 55/3, 55/4(P), 56/2(P), 56/3, 62/1, 62/2, 62/3(P), 62/4, 62/5, 62/6, 62/7, 63/2, 63/3, 63/4(P), 63/5 & 63/6, Gunjur Village, Varthur Hobli, Bangalore East Taluk, Bangalore Urban District
3	Number of Buildings	**3 High Rise Residential Buildings i.e.** Building-1 with 2 Wings i.e. Wing-A & Wing-B – joined together, Building-2 with 2 Wings i.e. Wing-C & Wing-D – joined together and Building-3 with 2 Wings i.e. Wing-E & Wing-F – joined together,
4	Number of floors	**Building-1** **Wing-A & Wing-B** Each of 01 common Basement, ground & 26 upper floors **Building-2** **Wing-C & Wing-D** Each of 2 common Basements, ground & 26 upper floors **Building-3** **Wing-E & Wing-F** Each of 2 common Basements, ground & 26 upper floors
5	Type of Occupancy **Part 4, Fire and Life Safety of Part-IV of NBC of 2016 clause 2.46** Occupancy or Use Group:— The principal occupancy for which a building or a part of a building is used or intended to be used; for the purpose of classification of a building according to the occupancy, an occupancy shall be deemed to include subsidiary occupancies which are contingent upon it. ***Part 4, Fire and Life Safety of Part-IV of NBC of 2016 clause 3.1.2 classification of residential buildings.*** ***3.1.2 Group A Residential Buildings*** These shall include any building in which sleeping accommodation is provided for normal residential purposes with or without cooking or dining or both	Residential- Group-A Sub Division-A4

	facilities, except any building classified under Group C. ***Subdivision A-4 Apartment houses*** — These shall include any building or structure in which living quarters are provided for three or more families, living independently of each other and with independent cooking facilities, for example, apartment houses, mansions and Chawls.	
6	Floor wise details of the occupancy :-	
		Building-1, 2 & 3 **Wing-A, B, C, D, E & F** Common Basement-1 For parking 695Cars, Fire Pump Room, 2 nosElectrical Panel Room, D.G. Room & STP, 2 nos utility panel room, roof water collection tank, UPS room, 2 nos communication room, 3 nos panel room, gents and ladies toilet, domestic pump room. **Building-2 & 3** **Wing-C, D, E & F** Common Basement-2 For parking 470 Cars, communication room, 2 nos panel room **Building-1** **Wing-A** Ground floor 4 Flats, Entrance lobby (double height) & Fire Command Center. 1 st floor 4 Flats & double height of Entrance lobby. 2 nd floor 4 Flats 3rd, 5th, 7th, 9th, 11 th , 13th, 15th, 17 th , 21 st , 23rd& 25 th floor 4 Flats on each floor x 11 floors = 44 Flats. 4th, 6th, 8 th , 10 th , 12 th , 14 th , 16 th , 18 th , 20 th , 22 nd 24th& 26 th floor 4 Flats on each floor x 12 floors = 48 Flats 19 th floor 4 Flats & Refuge area of 29.00 Sq. mtrs.

		Wing-B Ground floor 4 Flats, Entrance lobby (double height) & Fire Command Center. 1 st floor 4 Flats & double height of Entrance lobby. 2 nd floor 4 Flats 3rd, 5th, 7th, 9th, 11 th , 13th, 15th, 17 th , 21 st , 23rd& 25 th floor 4 Flats on each floor x 11 floors = 44 Flats. 4th, 6th, 8 th , 10 th , 12 th , 14 th , 16 th , 18 th , 20 th , 22 nd 24th& 26 th floor 4 Flats on each floor x 12 floors = 48 Flats 19 th floor 4 Flats & Refuge area of 39.00 Sq. mtrs. **Building-2** **Wing-C** Ground floor 4 Flats, Entrance lobby (double height) & Fire Command Center. 1 st floor 4 Flats & double height of Entrance lobby. 2 nd floor 4 Flats 3rd, 5th, 7th, 9th, 11 th , 13th, 15th, 17 th , 21 st , 23rd& 25 th floor 4 Flats on each floor x 11 floors = 44 Flats. 4th, 6th, 8 th , 10 th , 12 th , 14 th , 16 th , 18 th , 20 th , 22 nd 24th& 26 th floor 4 Flats on each floor x 12 floors = 48 Flats 19 th floor 4 Flats & Refuge area of 29.00 Sq. mtrs. **Wing-D**
	Basement Parking & services. (details to be provided)	

Ground floor

4 Flats, Entrance lobby (double height) & Fire Command Center.

1 st floor

4 Flats & double height of Entrance lobby.

2 nd floor

4 Flats

3rd, 5th, 7th, 9th, 11 th , 13th, 15th, 17 th , 21 st , 23rd& 25 th floor

4 Flats on each floor x 11 floors = 44 Flats.

4th, 6th, 8 th , 10 th , 12 th , 14 th , 16 th , 18 th , 20 th , 22 nd 24th& 26 th floor

4 Flats on each floor x 12 floors = 48 Flats

19 th floor

4 Flats & Refuge area of 39.00 Sq. mtrs

Building-3

Wing-E

Ground floor

4 Flats, Entrance lobby (double height) & Fire Command Center.

1 st floor

4 Flats & double height of Entrance lobby.

2 nd floor

4 Flats

3rd, 5th, 7th, 9th, 11 th , 13th, 15th, 17 th , 21 st , 23rd& 25 th floor

4 Flats on each floor x 11 floors = 44 Flats.

4th, 6th, 8 th , 10 th , 12 th , 14 th , 16 th , 18 th , 20 th , 22 nd 24th& 26 th floor

4 Flats on each floor x 12 floors = 48 Flats

19 th floor

4 Flats & Refuge area of 45.00 Sq. mtrs

Wing-F

Ground floor

		4 Flats, Entrance lobby (double height) & Fire Command Center. 1 st floor 4 Flats & double height of Entrance lobby. 2 nd floor 4 Flats 3rd, 5th, 7th, 9th, 11 th , 13th, 15th, 17 th , 21 st , 23rd& 25 th floor 4 Flats on each floor x 11 floors = 44 Flats. 4th, 6th, 8 th , 10 th , 12 th , 14 th , 16 th , 18 th , 20 th , 22 nd 24th& 26 th floor 4 Flats on each floor x 12 floors = 48 Flats 19 th floor 4 Flats & Refuge area of 45.00 Sq. mtrs _____ **Total - 648 Flats** _____
	Ground floor Details of occupancy. (details to be provided)	Mentioned Above.
	Upper floors No. of flats on each floor. (details to be provided)	Mentioned Above.
	Terrace floors Over head tanks, lift Machine room. (details to be provide)	Mentioned Above.
7	Height of the building (In Mtrs) **As per Part 3 Development Control Rules and General Building Requirements clause 2.10 of NBC 2016** **Building, Height of** –2.10 Building, Height of — The vertical distance measured in the case of flat roofs, from the average level of the ground around and contiguous to the building or as decided by the Authority to the terrace of last livable floor of the building adjacent to the external walls; and in the case of pitched roofs, up to the point where the external surface of the outer wall intersects the finished surface of the sloping roof; and in the case of gables facing the road, the mid-point between the eaves level and the ridge. Architectural features serving no other function except that of decoration shall be excluded for the purpose of measuring heights.	**Building-1, 2 & 3** **Wing-A, B, C, D, E & F** Each of 81.30 mtrs.
	Site Area (In Sq.Mtrs) **As per Part 3 Development Control Rules and**	

8	**General Building Requirements clause 2.75 of NBC Site (Plot)**— A parcel (piece) of land enclosed by definite boundaries.	90,244.90 Sq. mtrs.
9	Built up area of each floor. (Block wise)(In Sq.Mtrs)	**Building-1, 2 & 3** **Wing-A, B, C, D, E & F** Common Basement-1 26,405.10 Sq. mtrs. **Building-2 & 3** **Wing-C, D, E & F** Common Basement-2 16,642.10 Sq. mtrs. **Building-1** **Wing-A & Wing-B** Ground floor 1,332.30 Sq. mtrs. 1 st floor 1,183.40 Sq. mtrs. 2 nd floor to 18 th floor 22,026.90 Sq. mtrs. (1,295.70 Sq. mtrs. on each floor x 17 floors). 19 th floor 1,321.00 Sq. mtrs. 20 th floor to 26 th floor 9,069.90 Sq. mtrs. (1,295.70 Sq. mtrs. on each floor x 7 floors). Terrace floor 258.20 Sq. mtrs. **Building-2** **Wing-C & Wing-D** Ground floor 1,331.20 Sq. mtrs. 1 st floor 1,183.40 Sq. mtrs. 2 nd floor to 18 th floor 22,026.90 Sq. mtrs. (1,295.70 Sq. mtrs. on each floor x 17 floors). 19 th floor 1,321.00 Sq. mtrs.

		20 th floor to 26 th floor 9,069.90 Sq. mtrs. (1,295.70 Sq. mtrs. on each floor x 7 floors). Terrace floor 258.20 Sq. mtrs. **Building-3** **Wing-E & Wing-F** Ground floor 1,511.60 Sq. mtrs. 1 st floor 1,328.86 Sq. mtrs. 2 nd floor to 18 th floor 24,860.12 Sq. mtrs. (1,462.36 Sq. mtrs. on each floor x 17 floors). 19 th floor 1,509.44 Sq. mtrs. 20 th floor to 26 th floor 10,236.52 Sq. mtrs. (1,462.36 Sq. mtrs. on each floor x 7 floors). Terrace floor 260.00 Sq. mtrs.
10	Total Built-up area (In Sq.Mtrs)	1,53,136,66 Sq. mtrs.
11	Surrounding Properties	
	Front	North -Existing 17.80 mtrs. wide Gunjur – Chikkabellandur Road (Proposed to be widened to 18.00 mtrs.)
	Rear	South - Villas.
	Side 1	East - Agriculture land.
	Side 2	West - Agriculture land.
	B. Structural details indicating the fire prevention, fire fighting and evacuation measures to be indicated in the drawings	
	Width of the road to which the building abuts and whether it is hard surfaced to carry the weight of 45000 kgs. **As per Part 3 Development Control Rules and General Building Requirements clause 2.83 of NBC 2016** **Street :** Any means of access, namely, highway,	**Name of the Road :**The premises is abutting to Existing 17.80 mtrs.

1	street, lane, pathway, alley, stairway, passageway, carriageway, footway, square, place or bridge, whether a thoroughfare or not, over which the public have a right of passage or access or have passed and had access uninterruptedly for a specified period, whether existing or proposed in any scheme, and includes all bunds, channels, ditches, storm-water drains, culverts, footpaths, sidewalks, traffic islands, roadside trees and hedges, retaining walls, fences, barriers and railings within the street lines. **a. Name of the Road** **b. Width of the Road (mtrs)** **c. Type of Road(Asphalted or Kaccha road)** **d. Is road a Dead end**	wide Gunjur – Chikkabellandur Road (Proposed to be widened to 18.00 mtrs.), located on the northern side. **Width of the Road :**The premises is abutting to Existing 17.80 mtrs. wide Gunjur – Chikkabellandur Road (Proposed to be widened to 18.00 mtrs.), located on the northern side. **Type of Road : Asphalted road.** **Is road a Dead end : NO**
2	Number of entrances and width of each entrance to the premises & height clearance over the entrance. ***As per Part 3 Development Control Rules and General Building Requirements clause 4.6 (d)of NBC 2016*** 1) The main entrance to the plot shall be of adequate width to allow easy access to the fire engine and in no case shall it measure less than 6 m. 2) The entrance gate shall fold back against the compound wall of the premises, thus leaving the exterior access way within the plot free for movement of fire tender. 3) If the main entrance at the boundary wall is built over, the minimum clearance shall be 4.5 m. ***As per Part 3 Development Control Rules and General Building Requirements clause 4.6of NBC 2016*** a) The width of the main street on which the building abuts shall not be less than 12 m and one end of this street shall join another street not less than 12 m in width. b) The road shall not terminate in a dead end; except in the case of residential building, up to a height of 30 m. a. **Main entrance width (mtrs)** b. **Is Entrance gate provisioned** **c. Is any Parabola planned**	**Main Entrance width:**2 Entrance **, each** of 6.00 mtrs. width from Existing 17.80 mtrs. wide Gunjur – Chikkabellandur Road (Proposed to be widened to 18.00 mtrs.), located on the northern side. **Is Entrance gate provisioned : Yes** **Is any Pergola planned : NO**
	Width of open space (Setbacks) ***Width of open space (Setbacks) 24*** **As per Part 3 Development Control Rules and General Building Requirements of NBC 2016** **Clause 2.57 Open Space:** — An area, forming an integral part of the plot, left open to the sky. NOTE — The open space shall be the minimum distance measured	

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between the front, rear and side of the building and the respective plot boundaries.

2.58 Open Space, Front — An open space across the front of a plot between the building line and front boundary of the plot.

2.59 Open Space, Rear — An open space across the rear of a plot between the rear of the building and the rear boundary of the plot.

2.60 Open Space, Side — An open space across the side of the plot between the side of the building and the side boundary of the plot.

3	***As per Part 3 Development Control Rules and General Building Requirements clause of NBC 2016 Table 4 Side and Rear Open spaces to be left around the Building (Clause 8.2.3.1)*** ***Sl No.*** / ***Height of the Building*** / ***Side and rear open spaces to be left around the building*** 1. 10 3 2. 15 5 3. 18 6 4. 21 7 5. 24 8 6. 27 9 7. 30 10 8. 35 11 9. 40 12 10. 45 13 11. 50 14 12. 55 16 13. 70 17 14. 120 18 15. Above 120 20 ***NOTES***: **1** For buildings above 24 m in height, there shall be a minimum front open space of 6 m. **2.** Where rooms do not derive light and ventilation from the exterior open space, the width of such exterior open space as given in col 3 may be reduced by 1 m subject to a	**Height of the Building-1 (Wing-A & Wing-B) is 81.30 mtrs.** **Required Setback space is 16.00 mtrs. as per ZR.** **Allowed Setback is :-** **Wing-A & Wing-B - joined together** Front (North) : Minimum 17.16 mtrs. Rear (South) : Minimum 46.14 mtrs. Side (East) : Minimum 16.47 mtrs. Side (West) :Minimum 16.33 mtrs. **Height of the Building-2 (Wing-C & Wing-D) is 81.30 mtrs.** **Required Setback space is 16.00 mtrs. as per ZR.** **Allowed Setback is :-** **Wing-C & Wing-D - joined together** Front (North) : Minimum 27.31 mtrs. Rear (South) : Minimum 24.22 mtrs. Side (East) : Minimum 19.39 mtrs. Side (West) :Minimum 16.47 mtrs. **Height of the Building-3 (Wing-E & Wing-F) is 81.30 mtrs.** **Required Setback space is 16.00 mtrs. as per ZR.** **Allowed Setback is :-** **Wing-E & Wing-F - joined together** Front (North) : Minimum 53.34 mtrs.

	minimum of 3 m and a maximum of 8 m. No further projections shall be permitted. **3.** If the length or depth of the building exceeds 40 m, add to col (3) ten percent of length or depth of building minus 4.0 m subject to maximum requirement of 20 m. ***As per Part 3 Development Control Rules and General Building Requirements clauseof NBC 2016 Clause 4.6 (C):*** 1) The approach to the building and open spaces on all its sides shall be not less than 6 m in width, and a turning radius of minimum 9 m shall be provided for fire tender movement of fire tenders weighing up to 45 t. 2) The same shall be hard surface capable of taking the mass of fire tender, weighing up to 45 t minimum. For heavier fire tenders, the minimum width, turning radius and the hard surface capable of taking the fire tender loads shall be as per the requirement laid down by the Fire Department. The layout for the open space for fire tender movement shall be done in consultation with the Chief Fire Officer of the city, which shall be kept free of obstructions and shall be motorable. The compulsory open spaces around the building shall not be used for parking. 3) If the main entrance at the boundary wall is built over, the minimum clearance shall be 4.5 m. **a. Height of the building (mtrs)** **b. Setback space left (mtrs)** ***Note: Any specific claims on relaxation of setback to be notified with necessary supporting documents.***	Rear (South) : Minimum 17.58 mtrs. Side (East) : Minimum 73.66 mtrs. Side (West) :Minimum 19.39 mtrs. **Driveway space left :**8.00 mtrs. all around the each Building from the each Building line with a turning radius of minimum 9.00 mtrs. for the easy movement of fire vehicles.
4	Width of means of access ***As per Part 3 Development Control Rules and General Building Requirements of NBC 2016*** **Clause 4.3 Width of Means of Access** The residential plots shall abut on a public means of access like street/road which is 12mtrs wide. Plots which do not abut on a street/road shall abut/front on a means of access, the width and other requirements of which shall be as given in Table 1. ***Table 1 Width and Length of Means of Access (Clause 4.3)*** ***Sl no.*** | ***Width of means of access*** | ***Length of means of access*** **(1)** | **(2)** | **(3)** i. | 6.0 | 75 ii. | 7.5 | 150 iii. | 9.0 | 250	Street/ Road width:- The distance between Abutting Road & Buildings is 336.00 mtrs. and 12.00 mtrs and 8 mtrs wideinternal Road is available to reach the Buildings from the abutting the Road.

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iv.	12.0	400
v.	18.0	1000
vi.	24.0	Above 1000

Note: If the development is only on one side of the means of access, the prescribed widths may be reduced by 1 m in each case.

In no case, development on plots shall be permitted unless it is accessible by a public street of width not less than 6 m.

Street/ Road width (mtrs)

Arrangement for parking the cars and ramps.
Arrangement for parking the cars and ramps. 26

As per Part 3 Development Control Rules and General Building Requirements of NBC 2016

Clause 2.63: Parking Space — An area enclosed or unenclosed, covered or open, sufficient in size to park vehicles, together with a drive-way connecting the parking space with a street or alley and permitting ingress and egress of the vehicles.

Arrangement for parking the cars and ramps and conditions for buildings on podium.

4.6.1 Buildings on Podium

4.6.1.1 Podium is a horizontal projection (platform) extending beyond the building footprint on one or more sides, and may consist of one or more levels (see Fig. 8A).

4.6.1.2 Uses permitted Podium may be used for the following purposes:

a) Parking of vehicles . When used for parking, one WC, two urinals and two wash basins for every 500 cars or part thereof, shall be provided on each podium floor. At least one accessible toilet complying with the requirements given in B-9 shall be provided preferably near the accessible parking. Provision for driver.s rest room for non- residential building shall be made.

b) Fire and building services/utilities in accordance with the provisions of other Parts/ Sections of the Code.

c) Topmost podium slab which is open to sky maybe landscaped and/or be used as recreational open space; subject to provision of 1.6 m high parapet wall.

d) Other habitable uses may be allowed by counting it in FAR subject to light, ventilation and fire safety requirements.

Uses proposed in (a) to (c), shall not be counted towards FAR.

4.6.1.3 Requirements

Following requirements shall be satisfied for buildings

5	constructed on podium: a) A podium may be permitted in a plot of area 1 500 m2 or more. b) A podium, if provided with ramp, may be permitted in one or more levels, however the total height shall not exceed 30.0 m above ground level. c) In case a podium is not provided with ramp, but provided with car lift only, the same may also be permitted in one or more levels, however, the total height shall not exceed 9.0 m above ground level. d) Requirements for ramp for vehicles (see Fig. 8B): 1) One way ramp of clear width of minimum 3.0 m and two way ramp with clear width of minimum 6.0 m shall be provided for LMV. 2) One way ramp of clear width of minimum 4.5 m and two way ramp with clear width of minimum 9.0 m shall be provided for LCV. 3) One way ramp of clear width of minimum 6.0 m and two way ramp with clear width of minimum 12.0 m shall be provided for HMV. 4) Ramp slope shall be maximum 1 in 8. 5) After a 40 m length of continuous ramp, a flat surface of minimum 6.0 m length shall preferably be provided (see Fig. 8B). 6) If podium is accessible to fire tender, minimum 7.5 m wide ramp shall be required for fire engine access with maximum slope of 1 in 10. e) Podium shall not be permitted in required minimum front open space. f) Podium, if accessible to fire tender, shall be so designed so as to take the load of fire tender weighing up to 45 t minimum or as per the requirement laid down by the Fire Department. g) Requirement of accessibility for elders and persons with disabilities shall be ensured in compliance with the provisions of Annex B which may require providing ramps with specified gradient or accessible lifts for access to different levels. 4.6.1.4 Requirements for fire tender movement a) Buildings having height more than 15 m above ground level shall necessarily be accessible by fire tender, as follows (see Fig. 9A): 1) For buildings having floor area less than 10 000 m2, fire tenders shall have access to at least one-third of the perimeter of building which shall be minimum 6.0 m wide and having 9.0 m turning radius.	**Building-1, 2 & 3 (Wing-A, B, C, D, E & F)** Provision has been made to park 695 Cars at common Basement-1 parking area of Building-1, 2 & 3, 470 Cars at common Basement-2 parking area of Building-2 & 3 and 30 Cars on the open space available on the northern side, after leaving 8.00 mtrs. wide driveway from the Building line. **No. of ramps:** **Building-1, 2 & 3 (Wing-A, B, C, D, E & F)** 4 Ramps, 2 each of 6.00 mtrs. width & another 2, each of 4.5 mtrs. width for common Basement-1 and 2 Ramps, each of 4.50 mtrs. for common Basement-2 for the cars to reach each common Basement parking area of Building-1, 2 & 3. **Type of Ramp :**One way **Gradation = 1 : 10** **Location of Ramp** 2 Ramps, located on the northern side & another 2 Ramps, located on the western side for common Basement-1 and 2 ramps from the common Basement-1 to common Basement-2 are within the Building line..

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2) For buildings having floor area more than 10 000 m2, fire engine shall have an access to at least to half of the perimeter of building which shall be minimum 6.0 m wide and having 9.0 m turning radius.

b) If podium is not accessible by fire tender, the podium may be such that it is not extended beyond the building footprint to an extent more than 11.0 m on the side where fire tender access is provided (see Fig. 9B and Fig. 9C). Such restriction shall not apply in case podium is accessible by fire engine (see Fig. 9D).

c) Minimum 6.0 m driveway width and 9.0 m width at turning shall be available for fire tender movement all around the podium. NOTE . The width and turning radius of ramp for fire tender access, and requirements of motorable open space for fire tender movement given above pertain to fire tender weighing up to 45 t and its operability. For heavier fire tenders, these shall be as per the requirement laid down by the Fire Department [see also 4.6 (c)].

4.7 Cul-de-sacs giving access to plots and extending from 150 m to 275 m in length with an additional turning space at 150 m will be allowed only in residential areas, provided cul-de-sacs would be permissible only on straight roads and further provided the end of cul-de-sacs shall be higher in level than the level of the starting point of such dead end road. The turning space, in this case shall be not less than 81 m2 in area, with no dimension less than 9 m.

a. No. of ramps

b. Width of Ramp (mtr)

c.Type of Ramp; Single or double (one way/ two way)

d. Gradation

e. Location of Ramp

Staircases

As per NBC 2016, Part 4, Fire and Life Safetyclause 4.4.2.4.3 Staircases,

As mentioned in ***Part 4, Fire and Life Safety clause 1.2 All buildings, shall have a minimum of two staircases.***

The provisions of this Part are applicable to,

a) all high rise buildings; where any of these buildings have floor area more than 500 m^2 on any one or more floors;

6) Buildings with two basements or more, or with one basement of area more than 500 m^2 unless otherwise mentioned specifically in the provisions.

The minimum width of tread without nosing shall be 250 mm for staircase of residential buildings. The treads shall be constructed and maintained in a manner to prevent

6	slipping. The maximum height of riser shall be 150 mm. The number of risers shall be limited to 12 per flight. The staircases may be internal staircases or external staircases. *4.4.2.4.3.2 Internal staircases* The internal staircases may be constructed with an external wall, or otherwise, and shall comply with the following: a) Internal stairs shall be constructed of non- combustible materials throughout, and shall have fire resistant rating of minimum 120 min. b) A staircase shall not be arranged round a lift shaft. c) Exits shall not be used as a portion of a supply, return or exhaust air system serving adjoining areas. Any opening(s) shall not be permitted in walls or in doors, separating exits from adjoining areas. d) No flue chimney, electromechanical equipment, air conditioning units, gas piping or electrical panels shall be allowed in the stairway. e) Notwithstanding the detailed provision for exits in accordance with 4.2 and 4.3, ***the following minimum width shall be provided for staircases for respective occupancies:*** ***1) Residential (A-4) : 1.25 m.*** f) A handrail shall be provided on one side of the staircase of width less than 1 500 mm, and on both sides of the staircase of width 1 500mm and more. The projection of handrail(s) in the staircase width shall not be more than 115 mm. h) The design of staircase shall also take into account the following: 1) The minimum headroom in a passage under the landing of a staircase and under the staircase shall be 2.2 m 2) Access to exit staircase shall be through a fire door of a minimum 120 min fire resistance rating. 3) No living space, store or other fire risk shall open directly into staircases. 4) The exit (including staircases) shall be continuous from refuge floors or terrace level, as applicable, to the level of exit discharge. 5) No electrical shafts/air conditioning ducts or gas pipes, etc, shall pass through or open in the staircases. 6) Lifts shall not open in staircase. 7) No combustible material shall be used for decoration/wall panelling in the staircase. 8) Beams/columns and other building features shall not reduce the head room/ width of the staircase. 9) The floor indication board, indicating the	**Building-1, 2 & 3** **Wing-A, B, C, D, E & F** 12 Staircases (2 in each Wing) **Floor area** **Building-1** **Wing-A & B** Maximum 1,332.30 Sq. mtrs. per floor **Building-2** **Wing-C & D** Maximum 1,331.20 Sq. smtrs. per floor **Building-3** **Wing-E & F** Maximum 1,511.60 Sq. mtrs. per floor **Area of Basements** **Building-1, 2 & 3** **Wing-A,B,C, D, E & F** Common Basement-1 26,405.10 Sq. mtrs. Common Basement-2 16,642.10 Sq. mtrs. **No. of Basement s:** **Building-1, 2 & 3** **Wing-A,B, C, D, E & F** **Wing-A & B** **01 Basement** **Wing-C, D, E & F** **2 Basements.** **Fire Rating** : 120 min **Stairs around Lift** : NO **Stairs are clear from any other service routings** : YES **No other services is taken inside the stairs** : NO **Fire door rating** : 120 min. **Fire Signage board** : YES **Staircase terminated at Ground**

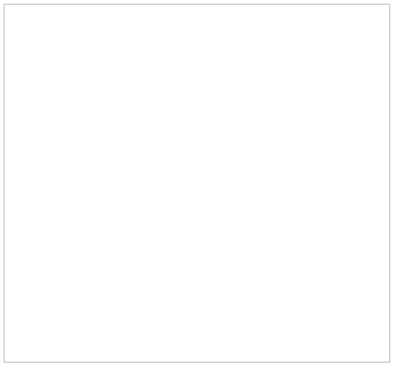
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	location/designated number of staircase, respective floor number and direction to exit discharge shall be placed inside the staircase, on the wall nearest to the fire door. It shall be of size not less than 300 mm × 200 mm (see Fig. 9). 10) Individual floors shall be prominently indicated on the wall outside the staircase and facing it. ***11) All staircases shall terminate at the level of exit discharge. The access to the basement shall be by a separate staircase.*** ***12) Scissors type staircases shall not be treated as part of exit.*** a. No. of Staircases in Blocks/Wings ***b.*** **Floor area (sqmt)** **c. Area of Basement (sqmt)** d. **No. of Basement** **e. Thread width (mm)** **f. No. of Risers (nos)** g. ***Fire Rating (min)*** h. ***Stairs around Lift : YES/ NO*** ***i. Stairs are clear from any other service routings : YES / NO*** ***j. No other services is taken inside the stairs : YES/ NO*** ***k. Staircase width : (mtrs)*** ***l. Staircase head room: (mtrs)*** ***m. Fire door rating : (min)*** ***n. Fire Signage board : YES/ NO*** ***o.*** **Staircase terminated at Ground level: YES/ NO**	**level** : **Building-1, 2 & 3** **Wing-A, B, C, D, E & F** All the Staircases are terminated at ground floor and 13 separate staircase have been proposed to reach common Basement-1 parking area and 9 separate staircases have been proposed to reach common Basement-2 parking area from the ground floor. Further provision has been made to enclose all the staircases at each floor of each Wing by using self closing smoke stop swing door at each floor level.
	EXTERNAL STAIRCASE ***4.4.2.4.3.4 External staircases*** The external staircases are the staircases provided on the external wall/facade, and shall comply with the following: a) External stairs shall always be kept in sound and usable condition. b) All external stairs shall be directly connected to the ground. c) Entrance to the external stairs shall be separate and remote from the internal staircase. d) Where an external staircase is provided, it shall be ensured that the use of it at the time of fire is not prejudiced by smoke and flame from openings (for example, windows, doors) in the external face of the building. Care shall be taken to ensure that no external wall or window	

7	opening opens on to or close to an external stair. If such openings exists within 3 m from an external staircase, they shall be protected with fire rated doors/window assemblies with rating of at least 60 min. e) The external stairs shall be constructed of non-combustible materials, and any doorway leading to it shall have minimum 120 min fire resistance. f) No external staircase shall be inclined at an angle greater than 45° from the horizontal. g) External stairs shall have straight flight not less than 1 500 mm wide. h) Handrails, to be provided on both sides, shall be of a height not less than 1 000 mm and not exceeding 1 200 mm. There shall be provisions of balusters with maximum gap of 150 mm. **a. Type of External staircase:** **b. Location of External stairs: YES/ NO** **c. Fire rating of door (min)** **d. Width of stairs: (mtrs)** **e. Section & Elevation of stairs:**	NO External Staircases.
8	Staircase Size:	
	Staircase Size	
	a. Width of the staircases. ***As per Clause 4.4.2.4.3.2 of Part 4 Fire and Life Safety of NBC 2016*** The following minimum width shall be provided for: ***1) Residential (A-4) : 1.25 m***	Building-1, 2 & 3 Wing-A to F Each of 1.25 mtrs. width
	b. Width of treads ***As per clause 4.4.2.4.3.1 of Part 4 Fire and Life Safety of NBC 2016:*** The minimum width of tread without nosing shall be 250 mm for staircase of residential buildings.	300 mm
	c. Height of riser ***As per clause 4.4.2.4.3.1 of Part 4 Fire and Life Safety of NBC 2016*** : The maximum height of riser shall be 150 mm for staircase of residential buildings.	150 mm
	d. Number of risers in a flight ***As per clause 4.4.2.4.3.1 of Part 4 Fire and Life Safety of NBC 2016:*** **T**he number of risers shall be limited to 12 per flight.	Maximum 10 Risers per flight.
	e. Height of hand rails ***As per clause 4.4.2.4.3.2 (f) of Part 4 Fire and Life Safety of NBC 2016:*** Handrails shall be provided at a height of 1 000 mm to be measured from the base of the middle of the treads to the top of the handrails. Balusters/railing shall be provided such that the width of staircase does not reduce.	1.00 mtr.

<table>
<tr><td></td><td>f. Head room clearance
As per Part 3 Development Control Rules and General Building Requirements clause 2.70 of NBC 2016

As per clause 4.4.2.4.3.2 (h) (1) of Part 4 Fire and Life Safety of NBC 2016:

The minimum headroom in a passage under the landing of a staircase and under the staircase shall be 2.2 m.</td><td>Minimum 2.20 mtrs.</td></tr>
<tr><td>9</td><td>Fire Tower
As per Part-4, NBC 2016, 2.24 Firefighting Shaft (Fire Tower) -An enclosed shaft having protected area of 120 min fire resistance rating comprising protected lobby, staircase and fireman's lift, connected directly to exit discharge or through exit passageway with 120 min fire resistant wall at the level of exit discharge to exit discharge. These shall also serve the purpose of exit requirement/ strategy for the occupants. The respective floors shall be approachable from fire-fighting shaft enabling the fire fighters to access the floor and also enabling the fire fighters to assist in evacuation through fireman's lift. The firefighting shall be equipped with 120 min fire doors. The firefighting shall be equipped with firemen talk back, wet riser and landing valve in its lobby, to fight fire by fire fighters (see Fig. 2 for a typical fire fighting shaft).
  </td><td><u>Building-1,, 2 & 3</u>

<u>Wing-A, B, C, D, E & F</u>

Proposed to provide Fire Tower in each Wing of each Building as per Part-4, NBC 2016, 2.24 Fire fighting Shaft (Fire Tower).

The Fire Tower lobby and Staircase in the Fire Fighting shaft are naturally/Cross ventilated, due to this fire tower may not be enclosed fire door is not provided for Fire Tower lift lobby as per NBC 2016.</td></tr>
<tr><td>10</td><td>Travel Distance
As per clause 2.59 of Part 4 Fire and Life Safety of NBC 2016:

The distance to be travelled from any point in a building to a protected exit or external escape route or final exit measured along the line of travel.

Table 5 Travel Distance (Based on Occupancy and Construction Type) (Clauses 4.4.2.1 and 4.4.2.2) of Part 4 0f NBC 2016.
<table><tr><td rowspan="2">Sl No.</td><td rowspan="2">Occupancy Group</td><td colspan="2">Maximum Travel distance</td></tr><tr><td>Type 1 & 2</td><td>Type 3 and</td></tr></table></td><td><u>Building-1, 2 & 3</u>

<u>Wing-A, B, C, D, E & F</u>

Maximum 44.94 mtrs. from the farthest point to staircases in common Basements.

Maximum 26.76 mtrs. from the farthest point and maximum 10.00 mtrs. from the dead end of the corridor to the staircases.</td></tr>
</table>

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		1 & 2	4
I.	***Residential (Group A)***	***30.00***	***22.50***

Notes:

1. For fully sprinklered building, the travel distance may be increased by 50 percent of the values specified.

2. Ramp shall not be counted as an exit in case of basement below the first basement in car parking.

Increased travel distance from the farthest point to staircases in Basement is acceptable as all the floors of all the Wings are proposed to be covered with automatic sprinkler system.

Number of lifts and capacity

Lift: An appliance designed to transport persons or materials between two or more levels in a vertical or substantially vertical direction by means of a guided car or a platform. The word elevator is also synonymously used for lift.

As per clause 4.15.1 of Part-4 Fire and Life Safety of NBC-2005

Where applicable, fire lifts shall be provided with a minimum capacity for 8 passengers and fully automated with emergency switch on ground level. In general, buildings 15 m in height or above shall be provided with fire lifts.

Fire Lifts — Following details shall apply for a fire lift:

1) To enable fire services personnel to reach the upper floors with the minimum delay, one fire lift per 1 200 m^2 of floor area shall be provided and shall be available for the exclusive use of the firemen in an emergency.

2) The lift shall have a floor area of not less than 1.4 m^2. It shall have loading capacity of not less than 545 kg (8 persons lift) with automatic closing doors of minimum 0.8 m width.

3) The electric supply shall be on a separate service from electric supply mains in a building and the cables run in a route safe from fire, that is, within the lift shaft. Lights and fans in the elevators having wooden panelling or sheet steel construction shall be operated on 24 V supply.

4) Fire fighting lift should be provided with a ceiling hatch for use in case of emergency, so that when the car gets stuck up, it shall be easily openable.

5) In case of failure of normal electric supply, it shall automatically trip over to alternate supply. For apartment houses, this changeover of supply could be done through manually operated changeover switch. Alternatively, the lift shall be so wired that in case of power failure, it comes down at the ground level and comes to stand-still with door open.

6) The operation of a fire lift is by a simple toggle or two-

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button switch situated in a glass-fronted box adjacent to the lift at the entrance level. When the switch is on, landing call-points will become inoperative and the lift will be on car control only or on a priority control device. When the switch is off, the lift will return to normal working. This lift can be used by the occupants in normal times.

7) The words 'Fire Lift' shall be conspicuously displayed in fluorescent paint on the lift landing doors at each floor level.

8) The speed of the fire lift shall be such that it can reach the top floor from ground level within 1 min.

Specification of lifts:

C-1.5 Lifts

General requirements of lifts shall be as follows:

11 a) Walls of lift enclosures shall have a fire rating of 2 h; lifts shafts shall have a vent at the top of area not less than 0.2 m^2.

b) Lift motor room shall be located preferably on top of the shaft and separated from the shaft by the floor of the room.

c) Landing doors in lift enclosures shall have a fire resistance of not less than 1 h.

d) The number of lifts in one row for a lift bank shall not exceed 4 and the total number of lifts in the bank (of two rows) shall not exceed 8. A wall of 2 h fire rating shall separate individual shafts in a bank.

e) Lift car door shall have a fire resistance rating of half an hour.

f) Collapsible gates shall not be permitted for lifts and shall have solid doors with fire resistance of at least 1 h.

g) If the lift shaft and lobby is in the core of the building, a positive pressure between 25 and 30 Pa shall be maintained in the lobby and a positive pressure of 50 Pa shall be maintained in the lift shaft. The mechanism for pressurization shall act automatically with the fire alarm; it shall be possible to operate this mechanically also.

h) Exit from the lift lobby, if located in the core of the building, shall be through a self closing smoke stop door of half an hour fire

resistance.

j) Lifts shall not normally communicate with the basement; if, however, lifts are in communication, the lift lobby of the basements shall be pressurized as in (g), with self-closing door as in (h).

k) Grounding switch(es), at ground floor level, shall be provided on all the lifts to enable the fire service to ground the lifts.

Building-1, 2 & 3

Wing-A, B, C, D, E & F

2 Passengers lifts, each of 13 passengers capacity & 1 Fire lift of 13 passengers capacity in each Wing (total 18 lifts)

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	m) Telephone or other communication facilities shall be provided in lift cars for building of 30 m in height and above. Communication system for lifts shall be connected to fire control room for the building. n) Suitable arrangements such as providing slope in the floor of lift lobby, shall be made to prevent water used during fire fighting, etc,at any landing from entering the lift shafts. p) A sign shall be posted and maintained on every floor at or near the lift indicating that in case of fire, occupants shall use the stairs unless instructed otherwise. The sign shall also contain a plan for each floor showing the locations of the stairways. Alternate source of power supply shall be provided for all the lifts through a manually operated changeover switch. No. Of Lifts Description Capacity	
12	Structural material RCC materials and brick walls of not less than two hours fire resistance should be used for the construction of structures. Only fire resistant materials or materials treated with fire retardant chemicals, should be used for interior decoration work. While attending the interior decoration the fixed fire fighting systems like sprinklers / risers etc., should not be covered or shifted from their original location.	**Structural material** RCC materials and brick walls of not less than two hours fire resistance should be used for the construction of structures. Only fire resistant materials or materials treated with fire retardant chemicals, should be used for interior decoration work. While attending the interior decoration the fixed fire fighting systems like sprinklers / risers etc., should not be covered or shifted from their original location.
	Basements: ***12.9.3. The basement shall have the following requirements:*** ***a) Every basement shall be in every part at least 2.4 m in height from the floor to the underside of the roof slab or ceiling;*** ***b) Adequate ventilation shall be provided for the basement. The ventilation requirements shall*** ***be the same as required by the particular occupancy according to byelaws. Any deficiency may be met by providing adequate mechanical ventilation in the form of blowers, exhaust fans, air conditioning systems, etc;*** ***c) The height of the ceiling of any basement shall be minimum 0.9 m and the***	

	maximum, 1.2 m above the average surrounding ground level. ***However, in case of parking, mercantile or business occupancy at ground floor, minimum height of the ceiling of the basement may be 0.3 m above the average surroundings ground level subject to mechanical ventilation being provided (see Fig. 11);***	
13	***d) Adequate arrangements shall be made such that surface drainage does not enter the basement;*** ***e) The walls and floors of the basement shall be watertight and be so designed that the effects of the surrounding soil and moisture, if any,*** ***are taken into account in design and adequate damp proofing treatment is given;*** ***f) The access to the basement shall be separate room the main and alternative staircase providing access and exit from higher floors.*** ***Where the staircase is continuous in the case of buildings served by more than one staircase, the same shall be of enclosed type serving as a fire separation from the basement floor and higher floors. Open ramps shall be permitted if they are constructed within the building line subject to the provision of (d);*** ***g) Access to basements through ramps shall be permitted subject to provision of (d). The requirements for the ramps shall be in accordance with 4.6.1.3 [see also Fig. 8 (b)];*** ***h) For all public buildings and uses including group housing, having basement going up to more than one level, access to all levels shall also be provided through lift. The exit requirements in basements shall comply with the provisions of Part 4 .Fire and Life Safety. of the Code.***	Basements shall be constructed as per Clause 12.9.3. of Part4 of NBC 2016 Proposed to provide Mechanical Ventilations for each common Basement parking area of Building-1, 2 & 3 as per NBC 2016.
	Smoke control of exits ***NBC 2016, Part 4, Fire and Life SafetyClause 4.4.2.5 Smoke control of exits*** ***a)*** *In building design, compartment-tation plays a vital part in limiting the spread of fire and smoke. The design should ensure avoidance of spread of smoke to adjacent spaces through the various leakage openings in the compartment enclosure, such as cracks, openings around pipes ducts, airflow grills and doors.*	

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In the absence of proper sealing of all these openings, smoke and toxic gases will obstruct the free movement of occupants of the building through the exits. Pressurization of staircases is of great importance for the exclusion of smoke and toxic gases from the protected exit.

b) Pressurization is a method adopted for protecting the exits from ingress of smoke, especially in high-rise buildings. In pressurization, air is injected into the staircases, lobbies, etc, as applicable, to raise their pressure slightly above the pressure in adjacent parts of the building. As a result, ingress of smoke or toxic gases into the exits will be prevented. The pressurization of staircases and lift lobbies shall be adopted as given in Table 6. The pressure difference for staircases shall be 50 Pa. Pressure differences for lobbies (or corridors) shall be between 25 Pa and 30 Pa. Further, the pressure differential for enclosed staircase adjacent to such lobby (or corridors) shall be 50 Pa. For enclosed staircases adjacent to non-pressurized lobby (or corridors), the pressure differential shall be 50 Pa.

c) Equipment and ductwork for staircase pressurization shall be in accordance with one of the following:

1) Directly connected to the stairway by ductwork enclosed in non-combustible construction.

2) If ducts used to pressurize the system are passed through shafts and grills are provided at each level, it shall be ensured that hot gases and smoke from the building cannot ingress into the staircases under any circumstances.

d) The normal air conditioning system and the pressurization system shall be designed and interfaced to meet the requirements of emergency services. When the emergency pressurization is brought into action, the following changes in the normal air conditioning system shall be effected:

1) Any re-circulation of air shall be stopped and all exhaust air vented to atmosphere.

2) Any air supply to the spaces/areas other than exits shall be stopped.

3) The exhaust system may be continued provided,

i) the positions of the extraction grills permit a general air flow away from the means of egress;

ii) the construction of the ductwork and fans is such that, it will not be rendered inoperable by hot gases and smoke; and

iii) there is no danger of spread of smoke to other floors by the path of the extraction system which can

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	be ensured by keeping the extraction fans running. ***e)*** *For pressurized stair enclosure systems, the activation of the systems shall be initiated by signalling from fire alarm panel.* *f) Pressurization system shall be integrated and supervised with the automatic/manual fire alarm system for actuation.* *g) Wherever pressurized staircase is to be connected to unpressurized area, the two areas shall be segregated by 120 min fire resistant wall.* *h) Fresh air intake for pressurization shall be away (at least 4 m) from any of the exhaust outlets/grille.* ***Pressurization of staircases & lift lobbies may be recommended as per requirement mentioned in Table-6.*** ***As per clause 2.49 of Part 4 Fire and Life Safety of NBC 2016:*** ***Pressurization*** *— The establishment of a pressure difference across a barrier to protect a* ***stairway, lobby, escape route*** *or room of a building from smoke penetration.* ***Smoke exhaust and Pressurization of areas above ground as per clause 4.6.1 of Part 4 Fire and Life Safety of NBC 2016*** *Corridors in exit access (exit access corridor) are created for meeting the requirement of use, privacy and layout in various occupancies. These are most often noted in hospitality, health care occupancies and sleeping accommodations.* *Exit access corridors of guest rooms and indoor patient department/areas having patients lacking self preservation and for sleeping accommodations such as apartments, custodial, penal and mental institutions, etc, shall be provided with 60 min fire resistant wall and 20 min self-closing fire doors along with all fire stop sealing of penetrations. Smoke exhaust system having make-up air and exhaust air system or alternatively pressurization system with supply air system for these exit access corridors shall be required. Smoke exhaust system having make-up air and exhaust air system shall also be required for theatres/auditoria. Such smoke exhaust system shall also be required for large lobbies and which have exit through staircase leading to exit discharge. This would enable eased exit of people through smoke controlled area to exit discharge.* *All exit passageway (from exit to exit discharge) shall be pressurized or naturally ventilated. The mechanical pressurization system shall be automatic in action with manual controls in addition. All such exit passageway*	Proposed to provide Natural ventilations at mid landing of each staircase and at corridor of each floor of each Wing as per Pat-4 of NBC 2016.. Proposed to provide smoke exhaust system with jet flans in each common Basement parking area of Building-1, 2 & 3 as per Part-4 of
14		

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shall be maintained with integrity for safe means of egress and evacuation. Doors provided in such exit passageway shall be fire rated doors of 120 min rating.

Smoke exhaust system where provided, for above areas and occupancies shall have a minimum of 12 air changes per hour smoke exhaust mechanism. Pressurization system where provided shall have a minimum pressure differential of 25-30 Pa in relationship to other areas.

The smoke exhaust fans in the mechanical ventilation system shall be fire rated, that is, 250°C for 120 min.

For naturally cross-ventilated corridors or corridors with operable windows, such smoke exhaust system or pressurization system will not be required.

Smoke Exhaust and Pressurization of areas below Ground.

As per clause 4.6.2 of Part 4 Fire and Life Safety of NBC 2016:

Each basement shall be separately ventilated. Vents with cross-sectional area (aggregate) not less than 2.5 percent of the floor area spread evenly round the perimeter of the basement shall be provided in the form of grills, or breakable stall board lights or pavement lights or by way of shafts.

Alternatively, a system of mechanical ventilation system may be provided with following requirements:

a) Mechanical ventilation system shall be designed to permit 12 air changes per hour in case of fire or distress call. However, for

normal operation, air changes schedule shall be as given in Part 8 'Building Services, Section 3 Air conditioning, Heating and Mechanical Ventilation' of the Code.

b) In multi-level basements, independent air intake and smoke exhaust shafts (masonry or reinforced concrete) for respective basement levels and compartments therein shall be planned with its make-up air and exhaust air fans located on the respective level and in the respective compartment. Alternatively, in multi-level basements, common intake masonry (or reinforced cement concrete) shaft may serve respective compartments aligned at all basement levels. Similarly, common smoke exhaust/outlet masonry (or reinforced cement concrete) shafts may also be planned to serve such compartments at all basement levels. All supply air and exhaust air fans on respective levels shall be installed in fire resisting room of 120 min. Exhaust fans at the respective levels shall be provided with back draft damper connection to the common smoke exhaust shaft ensuring complete isolation and

NBC 2016.

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compartmentation of floor isolation to eliminate spread of fire and smoke to the other compartments/floors.

c) Due consideration shall be taken for ensuring proper drainage of such shafts to avoid insanitation condition. Inlets and extracts may be terminated at ground level with stall board or pavement lights as before. Stall board and pavement lights should be in positions easily accessible to the fire brigade and clearly marked 'AIR INLET' or 'SMOKE OUTLET' with an indication of area served at or near the opening.

d) Smoke from any fire in the basement shall not obstruct any exit serving the ground and upper floors of the building.

e) The smoke exhaust fans in the mechanical ventilation system shall be fire rated, that is, 250°C for 120 min.

f) The smoke ventilation of the basement car parking areas shall be through provision of supply and exhaust air ducts duly installed with its supports and connected to supply air and exhaust fans. Alternatively, a system of impulse fans (jet fans) may be used for meeting the requirement of smoke ventilation complying with the following:

1) Structural aspects of beams and other down stands/services shall be taken care of in the planning and provision of the jet fans.

2) Fans shall be fire rated, that is, 250°C for 120 min.

3) Fans shall be adequately supported to enable operations for the duration as above.

4) Power supply panels for the fans shall be located in fire safe zone to ensure continuity of power supply.

5) Power supply cabling shall meet circuit integrity requirement in accordance with accepted standard [4(13)].

The smoke extraction system shall operate on actuation of flow switch actuation of sprinkler system. In addition, a local and/or remote 'manual start-stop control/switch' shall be provided for operations by the fire fighters. Visual indication of the operation status of the fans shall also be provided with the remote control. No system relating to smoke ventilation shall be allowed to interface or cross the transformer area, electrical switchboard, electrical rooms or exits. Smoke exhaust system having make-up air and exhaust air system for areas other than car parking shall be required for common areas and exit access corridor in basements/underground structures and shall be completely separate and independent of car parking areas and other mechanical areas. Supply air

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	shall not be less than 5 m from any exhaust discharge openings.	
15	Compartmentation ***Compartmentation 42*** ***As per clause 4.5 of Part 4 Fire and Life Safety of NBC 2016:*** 4.5.1 General a) It is important to limit the spread of a fire in any building. The usual method is to use fire barriers. In some instances these barriers need to be penetrated for ductwork, plumbing and electrical systems, and in such cases, use of passive fire protection measures shall be done so that the integrity of these barriers is not compromised. b) Floor(s) shall be compartmented with area as given below. 4.5.2 All floors shall be compartmented / zoned with area of each compartment being not more than 750 m^2. The maximum size of the compartment shall be as follows, in case of sprinklered Basement/Building: ***Sl No / Use / Compartment-ation Area m2*** 1. *Basement car parking* — *3000* 2. *Basements (other than car parking)* — *2000* ***In addition, there shall be requirement of a minimum of two compartments if the floor plate size is equal or less than the areas mentioned above. However, such requirement of minimum two compartments shall not be required, if the floor plate is less than 750 m^2. Compartmentation shall be achieved by means of fire barrier having fire resistance rating of 120 min.***	Proposed to provide Water curtain system with K-23 nozzles in each common Basement parking area of Building-1, 2 & 3 as per NBC-2016.
	Gas Supply ***Gas Supply*** ***As per clause 4.7.1 of Part 4 Fire and Life Safety of NBC 2016:*** ***Town Gas/ LPG supply pipes*** Where gas pipes are run in buildings, the same shall be run in separate shafts exclusively for this purpose and these shall be on external walls, away from the staircases. Gas distribution pipes shall always be below the false ceiling. The length of these pipes shall be as short as possible. In the case of kitchen cooking range area, hood should have grease filters using metallic grill to trap oil vapours escaping	

16	into the fume hood. NOTE — For detailed information on gas pipe installations, reference may be made to Part 9 'Plumbing Services, Section 4 Gas Supply' of the Code. 4.7.2 Thermal detectors These shall be installed into fume hoods of large kitchens for hotels, hospitals, and similar areas located in high rise buildings. Arrangements shall be made for automatic tripping of the exhaust fan in case of fire. If gas is used, the same shall be shut off. The voltage shall be 24 V or 100 V d.c. operated with external rectifier. The valve shall be of the hand re-set type and shall be located in an area segregated from cooking ranges. Valves shall be easily accessible. The hood shall have manual facility for steam or suitable hood extinguishing gas released depending on duty condition. **4.7.3** Gas cylinders and manifold shall need to be housed in a detached location with no other occupancy within distances prescribed in good practice [4(14)] thereof. There shall be an enclosure suitably ventilated. It is desirable to provide medium velocity spray nozzles which can be operated by quick opening valve situated away from the enclosure. **4.7.4** In the case of gas cylinders, if manifold has to be installed on podium/close to podium, the same shall be away from any air intakes/smoke exhaust openings/ any windows. **4.7.6** Gas meters shall be housed in a suitably constructed metal cupboard located in a well-ventilated space, keeping in view the fact that LPG is heavier than air and town gas is lighter than air. **4.7.7** Wherever LPG reticulation/cylinders are used in buildings above 100 m, gas leak detectors shall be provided at the usage points and monitored from fire command centre. The cables used for signalling shall be circuit integrity cables. **4.7.8** The gas lines shall not be installed through any electrical shafts, escape routes, refuge areas / refuge floors. **4.7.9** Kitchens working on LPG fuel shall not be permitted in basements.	As per drawings not proposed. If Town gas / LPG supply systems is proposed in the building, it shall be provided as per clause 4.7.1 of Part 4 Fire and Life Safety of NBC 2016& Separate NOC has to be obtained from this department
	3.4.5.4 Service ducts and shaft Openings in walls or floors which are necessary to be provided to allow passages of all building services like cables, electrical wirings, telephone cables, plumbing pipes, etc, shall be protected by enclosure in the form of ducts/shafts having a fire resistance not less than 120 min. The inspection door for electrical shafts/ducts low voltage wiring running in shafts/ducts, shall either be armoured type or run through metal conduits. The space between the electrical cables/conduits and the walls/slabs shall be filled in by a fire stop material having fire resistance rating of not less than 120 min. This shall exclude requirement of fire stop sealing for low voltage services shaft.	

17	For plumbing shafts in the core of the building, with shaft door opening inside the building, the shafts shall have inspection doors having fire resistance rating not less than 30 min. For plumbing shafts doors which open in wet areas or in naturally ventilated areas or on external wall of the building, the shafts may not require doors having any specified fire rating. ***3.4.6 Electrical Installation*** 3.4.6.1 The electric distribution cables/wiring shall be laid in a separate shaft. The shaft shall be sealed at every floor with fire stop materials having the same fire resistance as that of the floor. High, medium and low voltage wiring running in shaft and in false ceiling shall run in separate shaft/conduits. Water mains, gas pipes, telephone lines, intercom lines or any other service line shall not be laid in the duct for electrical cables; use of bus ducts/solid rising mains instead of cables is preferred.	Proposed to provide Service ducts and shafts and provision has been made to seal the ducts at each floor level of each Wing as per specifications.
	Escape Lighting and Exit Signage's. ***3.4.7 Escape Lighting and Exit Signage*** Exit access, exits and exit discharge shall be properly identified, with adequate lighting maintained in the elements of the egress systems so that all occupants shall be able to leave the facility safely. ***3.4.7.1 Lighting*** a) The exit, exit access and exit discharge systems shall be illuminated continuously. The floors of the means of egress shall be illuminated at all points, including angles and intersections, in corridors and passageways, stairwells, landings of stairwells and exit. b) Emergency lighting shall be powered from a source independent of that supplying the normal lighting. c) Escape lighting shall be capable of, 1) indicating clearly and unambiguously the escape routes; 2) providing adequate illumination along such routes to allow safe movement of persons towards and through the exits; and 3) Ensuring that fire alarm call points and Fire fighting equipment provided along the escape routes can be readily located. d) The horizontal luminance at floor level on the centreline of an escape route shall not be less than 10 lumen / m^2. In addition, for escape routes up to 2 m wide, 50 percent of the route width shall be lit to a minimum of 5 lumen / m^2. e) Required illumination shall be arranged such that the failure of any single lighting unit, such as the burning out of one luminaire, will not leave any area in darkness and does not impede the functioning of the system further.	

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18	f) The emergency lighting shall be provided to be put on within 5 s of the failure of the normal lighting supply. Also, emergency lighting shall be able to maintain the required illumination level for a period of not less than 90 min in the event of failure of the normal lighting even for smaller premises. g) Battery pack emergency lighting, because of its limited duration and reliability, shall not be allowed to be used in lieu of a diesel engine driven emergency power supply. h) Escape lighting luminaries should be sited to cover the following locations: 1) Near each intersection of corridors, 2) At exits and at each exit door, 3) Near each change of direction in the escape route, 4) Near each staircase so that each flight of stairs receives direct light, 5) Near any other change of floor level, 6) Outside each final exit and close to it, 7) Near each fire alarm call point, 8) Near fire fighting equipment, and 9) To illuminate exit and safety signs as required by the enforcing authority. ***NOTE***. For the purpose of this clause 'near' is normally considered to be within 2 m measured horizontally. j) The luminaries shall be mounted as low as Possible, but at least 2 m above the floor level. k) Signs are required at all exits, emergency exits and escape routes, which should comply with the graphic requirements of the relevant Indian Standards. ***3.4.7.2*** Exit passageway (at ground) and staircase lighting shall also be connected to alternative supply. The alternative source of supply may be provided by battery continuously trickle charged from the electric mains. ***3.4.7.3*** Suitable arrangements shall be made by installing double throw switches to ensure that the lighting installed in the staircase and the corridor does not get connected to two sources of supply simultaneously. Double throw switch shall be installed in the service room for terminating the stand-by supply. The emergency lighting system shall be well maintained by periodical inspections and tests so as to ensure their perfect serviceability at all times. ***3.4.7.4*** Exit signage Where exit access is provided through corridors / paths, the occupants shall be able to easily identify the way to exits. Exit signs shall be provided such that no point in an exit access is more than 30 m from a	Escape lighting and Exit signage's should be provided as per clause 3.4.7, Lighting 3.4.7.1

	visible exit directional sign. An exit sign indicating the direction to an exit shall be provided at all changes in direction. Exits shall be clearly visible and the route to reach the exits shall be clearly marked and signs posted to guide the occupants of the floor concerned. Signs shall be illuminated and wired to an independent electrical circuit on an alternative source of supply. The sizes and colours of the exit signs shall be in accordance with good practice [4(7)]. The colour of the exit signs shall be green. ***NOTE.*** This provision shall not apply to A-2 and A-4 occupancies less than 15 m in height. The exit sign with arrow indicating the way to the escape route shall be provided at a suitable height from the floor level on the wall and shall be illuminated by electric light connected to corridor circuits. All exit way marking signs should be so installed that no mechanical damage shall occur to them due to moving of furniture or other heavy equipment. Further, all landings of floor shall have floor indicating boards prominently indicating the number of the floor. Photo luminescent markings shall be pasted at internal hydrant boxes.	
	D. The builder should arrange for the following fire fighting and evacuation measures:-	
1	Electric Power Supply ***NBC 2016, Part-4 Fire and Life Safety 3.4.6.2*** Emergency power for fire and life safety systems Emergency power supplying distribution system for critical requirement for functioning of fire and life safety system and equipment shall be planned for efficient and reliable power and control supply to the following systems and equipment where provided: a) Fire pumps. b) Pressurization and smoke venting; including its ancillary systems such as dampers and actuators. c) Fireman's lifts (including all lifts). d) Exit signage lighting. e) Emergency lighting. f) Fire alarm system. g) Public address (PA) system (relating to Emergency voice evacuation and annunciation). h) Magnetic door hold open devices. j) Lighting in fire command centre and security room. Power supply to these systems and equipment shall be from normal and emergency (standby generator) power sources with changeover facility. If power supply, is from HV source and HV generation, the transformer should be	Proposed to provide 3 Nos. Diesel generators, each of 500 KVA capacity at common Basement-1 of Building-1, 2 & 3 to provide service

	planned in standby capacity to ensure continuity of power to such systems. Wherever and backup DG sets are of higher voltage rating, then dual redundant cables shall be taken to all transformers. The generator shall be capable of taking starting current of all the fire and life safety systems and equipment as above. Where parallel HV/LV supply from a separate substation fed from different grid is provided with appropriate transformer for emergency, the provision of generator may be waived in consultation with the Authority. ***3.4.6.4 Standby supply*** ***Diesel generator set(s) shall not be installed at any floor other than ground/first basement.*** If the same are installed indoors, proper ventilation and exhaust shall be planned. The DG set room shall be separated by 120 min fire resistance rated walls and doors. The oil tank for the DG sets (if not in the base of the DG) shall be provided with a dyked enclosure having a volumetric capacity of at least 10 percent more than the volume of the oil tank. The enclosure shall be filled with sand for a height of 300 mm. For detailed information regarding fire safety requirements for hazardous petroleum products, reference may be made to The Petroleum Act, 1934 and the Rules framed there under.	to all the emergency provisions in the Buildings.
2	Down comer system ***Down comer system.*** *47* **NBC-2016, Part-4, Fire & Life Safety, Down-comer** — An arrangement of fire fighting within the building by means of down-comer pipe connected to terrace tank through terrace pump, gate valve and non-return valve and having mains not less than 100 mm internal diameter with landing valves on each floor/landing. It is also fitted with inlet connections at ground level for charging with water by pumping from fire service appliances and air release valve at roof level to release trapped air inside. ***NBC 2016, Part-4, Fire & Life Safety Table 7 (6) down comer shall be provided for every 1000 sq.mtrs. built up area,*** ***Apartment Houses (A-4)*** 1. 1. For 15 m and above but not exceeding 35 m in height. The down comer should be of 100 mm internal diameter and G.I. 'C' class pipe. From each down comer single hydrant outlet should be provided	Not Required.
	Wet riser system **NBC 2016 Part-4, Fire & Life Safety, Clause 2.65 Wet Riser** — An arrangement for fire fighting within the building by means of vertical rising mains not less than 100 mm nominal diameter with landing valves on each floor/landing for fire fighting purposes and permanently	**Building-1, 2 & 3** **Wing-A, B, C, D, E & F**

3	charged with water from a pressurized supply. ***NBC -2016, Part-4 Fire & Life Safety, Table 7 (5) wet riser shall be provided for every 1000 sq.mtrs. built up area,*** ***Apartment Houses (A-4)*** 1. 1. 35 m and above in height. The riser should be of 100 mm internal diameter and G.I. 'C' class pipe. From each riser single hydrant outlet should be provided at each landing	Proposed to provide 6 Wet Riser-cum down comer systems (01 in each Wing) near the staircases. Each Riser will be of 100 mm internal diameter and will be of G.I. 'C' class pipe. From each riser single hydrant outlet at each floor landing will be provided.
4	Hose reel hose system. ***NBC-2016, Part-4, Fire and Life Safety, Table 7 (4) First Aid Hose reel shall be provided for,*** ***Apartment Houses (A-4)*** 1. Should be provided in all the buildings irrespective of height and irrespective of occupancy. Rubber lined Hose reel hose of size minimum 19 mm of 40 mtr length as per IS 884, with Gate valve (upstream) and shut off nozzle of 5 mm size. The hose reel hose should be connected at each landing by means of an adaptor. Adequate BIS marked re-inforced rubber lined delivery hoses of 63 mm size to reach the farthest point of the floor / setbacks from the system should be provided with a branch pipe near each hydrant outlet in a proper box to protect it from withering	Hose reel hose of 40 mtrs. length with drum and 2 Nos. delivery hose pipes, each of 15 mtrs. length with gunmetal branch pipe will be provided inside the hose cabinet near each outlet of each Tower & Club House. .
5	Yard hydrant system. **NBC-2016, Part-4, Clause 2.64.1 : Hydrant system**– A distribution system having a network of piping installed underground / above ground around and / or through inside of a building with internal and / or external hydrants fitted with landing walls at regular interval according to the occupancy. The distribution system is connected to water supply system from fire fighting. ***NBC-2016, Part-4, Table 7 (7)Yard hydrant shall be provided for,*** ***Apartment Houses (A-4)*** 1. 45 m and above in height. At least two fire service inlets to boost the water in the riser directly from the mobile pump should also be provided. These inlets should be located at an easily accessible position, preferable near the entry point to the premises.	Proposed to provide 17Nos. Yard hydrants around the Buildings and 01 No. 04 Way & 01 No. 2 way Fire Service inlets near the entrances.
	Underground static water storage tank combined capacity for wet riser, yard hydrant and sprinklers per set of pumps ***NBC-2016, Part-4, Table 7 (11) Underground Static Water Storage Tank Combined Capacity for Wet***	

	Riser, Yard hydrant and Sprinklers per set of Pumps shall be provided for, ***Apartment Houses (A-4)*** 1. Above 35 m but not exceeding 45 m. In height – 75,000 lts. 2. Above 45 m but not exceeding 60 m. In height – 1,50,000 lts. 3. Above 60 m in height. – 2,00,000 lts. ***Note: Fire tank to be always filled with water. Over flow of fire tank to be taken to domestic tank. Arrangement should be such that any incoming water should first fill-up fire tank, then overflow to other utilizations.***	
6	*H-4 ENCLOSED PARKING STRUCTURES* c) For basement car parking, compartmentation can be achieved, with fire barrier or with water curtain nozzle (K-23) or with combination thereof. Automatic deluge system comprising deluge valve, piping, nozzles, etc shall be used to zone the compartment in case of water curtain system. In case of water curtain, existing water storage shall be supplemented by water demand for water curtain nozzles for 60 min considering the largest compartment. perimeter out of all compartments of car parking in any of the basements. d) The water supply for the water curtain nozzles shall be through independent electric pump of adequate capacity (flow and head) with piping/riser for the water supply to the nozzles. e) The water curtain shall be operated by the actuation of flow switch actuating sprinkler system. f) For smoke ventilation requirement of car parking, see 4.6.2. g) All fire exit doors from the car parking to exits shall be painted green and shall display exit signage.	The Wet Riser-cum-down comer systems of Building-1 to 3 (Wing A to F) will be connected to 2 Nos. underground tanks, each of 2,00,000 liters capacity (total 4,00,000 litres .capacity). Similarly the Water curtain systems will be connected to one separate underground tank of 2,00,000liters capacity.
7	Terrace tank ***NBC-2016, Part-4 Table 7(12) Terrace Tank Over Respective Tower Terrace shall be provided for*** ***Apartment Houses (A-4)*** 1. Less than 15 m in height 5000 lts. (5000 lts.) **(Note 6 :** Additional value given in parenthesis shall be added if basement area exceeds 200 m^2) 2. 15 m and above but not exceeding 35 m in height 25,000 lts. 3. Above 35 m but not exceeding 45 m in height 5000 lts. 4. 45 m and above in height 10,000 lts. **Note: Over head tank to overflow to domestic tank.** ***Arrangement should be such that any incoming***	Each Wet riser-cum-down comer system of each Wing of each Building will also be connected to an overhead tank of 10,000 liters capacity. (total 06 overhead tanks).

	water should first fill-up fire tank, then overflow to other utilizations.	
8	Pump near underground static water storage tank (fire pump) with minimum pressure of 3.5 kg/cm2 at terrace level ***NBC-2016, Part-4, Table 7 (13) Pump near underground static water storage tank (Fire pump) with minimum pressure of 3.5 kg/cm2 at remotest location.*** ***Apartment Houses (A-4)*** 1. Above 35 m but not exceeding 45 m in height (***Note 10*** **:** One electric and one diesel pump of capacity 2280 l/min and one electric pump of capacity 180 l/min. See also Note 22 and 23) *(**Note 22***: One set of pumps shall be provided for each 100 hydrants or part thereof, with a maximum of two sets. In case of more than one pump set installation, both pump sets shall be interconnected at their delivery headers. (***Note 23***: Alternative to provisions of additional set of pumps, the objective can be met by providing additional diesel pump of the same capacity and doubling the water tank capacity as required for one set of pumps.) 2. Above 45 m in height but not exceeding 60 m in height (***Note 11*** Provide required number of sets of pumps each consisting of two electric and one diesel pump (stand by) of capacity 2 280 litre/min and two electric pump of capacity 180 litre/min *(see* Fig. 12) *(see also* Notes 22 and 23). (***Note 22*** One set of pumps shall be provided for each 100 hydrants or part thereof, with a maximum of two sets. In case of more than one pump set installation, both pump sets shall be interconnected at their delivery headers. ***Note 23*** Alternative to provisions of additional set of pumps, the objective can be met by providing additional diesel pump of the same capacity and doubling the water tank capacity as required for one set of pumps.) 1. Above 60 m in height (***Note 12*** Provide required number of sets of pumps each consisting of two electric and one diesel pump (stand by) of capacity 2 850 litre/min and two electric pump of capacity 180 litre/min *(see* Fig. 12) *(see also* Notes 22 and 23) (***Note 13*** Lower levels in high rise buildings 60 m or above in height are likely to experience high pressure and therefore, it is recommended to consider multi-stage, multi-outlet pumps (creating pressure zones) or variable frequency drive pumps or any other equivalent	The Wet riser-cum-down comer systems of Building-1 to 3 (Wing-A to F) will be connected to 2 sets of pump I,e 4 Nos. of end suction pumps with orifice plate for high pressure reduction Electrical driven pumps (2 nos for hydrant & 2 nos for sprinkler) and 2 Nos. of end suction pumps with orifice plate for high pressure reduction plate for high pressure reduction Diesel Diesel driven pumps, each capable of delivering 2850 litres of water per minute and 4 Nos. jockey pumps, each capable of delivering 180 litres of water per minute at common Pump Room. Further the Water Curtain system will be connected to an separate Electrically driven pump, capable of delivering 2850 litres of water per minute (flow and head) with piping/riser for the water supply to the nozzles.

	arrangement) (***Note 22*** One set of pumps shall be provided for each 100 hydrants or part thereof, with a maximum of two sets. In case of more than one pump set installation, both pump sets shall be interconnected at their delivery headers. (***Note 23*** Alternative to provisions of additional set of pumps, the objective can be met by providing additional diesel pump of the same capacity and doubling the water tank capacity as required for one set of pumps.) ***If Basement is compartmented using water curtains additional pump as per clause H-4 d) The water supply for the water curtain nozzles shall be through independent electric pump of adequate capacity (flow and head) with piping/riser for the water supply to the nozzles to be provided.***	
9	Pumps at the Terrace Tank level with Minimum Pressure of 2.0 kg/cm2. ***NBC-2016, Part-4, Table 7 (14) pumps at the Terrace tank level with Minimum Pressure of 3.5 kg/cm2 shall be provided for,*** ***Apartment Houses (A-4)*** 1. Less than 15 m in height 450 LPM (450 LPM) (***Note 6***: Additional value given in parenthesis shall be added if basement area exceeds 200 m^2.) 2. 15 m and above but not exceeding 35 m in height – 900 LPM.	Not required.
10	Manually operated fire alarm system. **NBC-2016, Part-4, Clause 2.1 Alarm System** —Fire alarm system comprising components for automatically detecting a fire, initiating an alarm of fire and initiating other actions as appropriate. NOTE — The system may also include manual fire alarm call points. ***NBC-2016, Part-4, Table 7 (9) Manually operated Electric Fire alarm system is required*** **Apartment Houses (A-4)** 1. 15 m and above in height. Manually operated electrical fire alarm system should be installed with call boxes located near each staircase landing of each building. The call boxes should be of 'break glass' type, where the call is transmitted automatically to the control room when the glass of the system is broken. This system should also be connected to an alternative source of power supply (diesel generator). The call boxes should be so installed that their location can be easily noticed from either direction and should be at a height of one meter from the floor level.	Proposed to provide manually operated Electrical Fire Alarm System with call point near each staircase landing at each floor of each Wing of each Building and its control panel at ground floor of each Wing

Proposed to provide Automatic fire detection with smoke detector heads & heat detector heads at each floor of each Wing of each Building as indicated below ;-

Floor	**Smoke `detector heads**	**Heat detector heads**
Building-1, 2 & 3		
Wing-A, B, C, D, E & F		
Common Basement-2	12	08
Common Basement-1	18	32
Building-1 Wing-A		--
Ground floor	17	04
1 st floor	20	04
2 nd floor	19	04
3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd& 25 th floor	19 on each floor (Total – 209 Nos.)	04 on each floor (Total – 44 Nos.)
19 th floor	18	04
4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26 th floor	19 on each floor (Total – 228 Nos.)	04 on each floor (Total - 48 Nos.)
Wing-B		
Ground floor	18	04
1 st floor	22	04
2 nd floor	20	04
3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd&	20 on each floor (Total 220 Nos.)	04 on each floor (Total - 44 Nos.

			25 th floor		
			19 th floor	19	04
			4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26 th floor	20 on each floor (Total – 240 Nos.)	04 on each floor (Total – 48 Nos.)
			Building-2		
			Wing-C		
			Ground floor	17	04
			1 st floor	20	04
			2 nd floor	19	04
			3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd& 25 th floor	19 on each floor (Total – 209 Nos.	04 on each floor (Total – 44 Nos.
			19 th floor	18	04
			4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26 th floor	19 on each floor (Total – 228 Nos.)	04 on each floor (Total – 48 Nos.)
			Wing-D		
			Ground floor	18	04
			1 st floor	22	04
			2 nd floor	20	04
			3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd& 25 th floor	20 on each floor (Total – 220 Nos.	04 on each floor (Total 44 Nos.
			19 th floor	19	04
			4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26 th floor	20 on each floor (Total - 240 Nos.	04 on each floor (Total – 48 Nos.
11	Automatic fire detection system and alarm system **NBC -2016, Part-4, Clause 2.1** **Automatic Fire Detection system with smoke detectors & heat detectors is required.** — Fire alarm system comprising components for automatically detecting a fire, initiating an alarm of fire and initiating other actions as appropriate. ***1.As per Table 7(10) Automatic Detection and Alarm System.*** ***Apartment Houses (A-4).*** Above 60 m in height (Automatic detection and alarm system is not required to be provided in car parking area. Such detection system shall however be required in other areas of car parking such as electrical rooms, cabins and other areas)				

		Building-3 **Wing-E** Ground floor: 24 / 04 1st floor: 28 / 04 2nd floor: 26 / 04 3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd& 25th floor: 26 on each floor (Total - 286 Nos.) / 04 on each floor (Total – 44 Nos.) 19th floor: 25 / 04 4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26th floor floor: 26 on each floor (Total – 312 Nos. / 04 on each floor (Total – 48 Nos.) **Wing-F** Ground floor: 24 / 04 1st floor: 28 / 04 2nd floor: 26 / 04 3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd& 25th floor: 26 on each floor (Total – 286 Nos.) / 04 on each floor (Total – 44 Nos. 19th floor: 25 / 04 4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26th floor: 26 on each floor (Total – 312 Nos.) / 04 on each floor (Total – 48 Nos.)
12	Public Address System *A system of two way talk back speaker with push to talk speakers to be provided at every staircase or firemen telephone to be provided at every staircase. Necessary console & amplifier with micro phone to be provided at ground floor in fire command center.*	Proposed to provide Public Address System with two way communication facility near each staircase landing at each floor of each Wing of each Building and its console at ground floor of each Wing.
		Proposed to provide Automatic Sprinkler system with sprinkler

heads and Water curtain system withK-23 Nozzles as indicated below:-

Floor	Sprinkler Heads	Water curtain Nozzles
Building-1, 2 & 3		
Wing-A, B, C, D, E & F		
Common Basement-2	1,420	150
Common Basement-1	2056	227
Building-1		
Wing-A		
Ground floor	32	--
1 st floor	39	--
2 nd floor	34	--
3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd& 25 th floor	32 on each floor (Total – 352 Nos.)	--
19 th floor	40	--
4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26 th floor	34 on each floor (Total – 408 Nos.)	--
Wing-B		--
Ground floor	34	--
1 st floor	43	--
2 nd floor	36	--
3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd& 25 th floor	36 on each floor (Total – 396 Nos.)	--
19 th floor	40	
4th, 6th,	36 on	--

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		8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26 th floor	each floor (Total – 432Nos.)	
		Building-2		--
		Wing-C		
		Ground floor	32	--
		1 st floor	39	--
		2 nd floor	34	--
		3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd& 25 th floor	32 on each floor (Total – 352 Nos.)	--
		19 th floor	40	--
		4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26 th floor	34 on each floor (Total – 408 Nos.)	--
		Wing-D		
		Ground floor	34	--
		1 st floor	43	--
		2 nd floor	36	--
		3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd& 25 th floor	36 on each floor (total – 396 Nos.)	--
		19 th floor	40	--
		4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26 th floor	36 on each floor (total 432 Nos.)	
		Building-3		
		Wing-E		
		Ground	42	--

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Automatic sprinkler system

Automatic Sprinkler System —A system of water pipes fitted with sprinkler heads at suitable intervals and heights and designed to actuate automatically, control and extinguish a fire by the discharge of water.

NBC-2016, Part-4, Table 7 (8) Automatic Sprinkler system.

Apartment Houses (A-4).

1. Upto 35 m in height.

(***Note 4*:** Required to be installed in basement if area of basement exceeds 200 m^2)

2. Above 35 m but not exceeding 45 m in height.

(***Note 4*:** Required to be installed in basement if area of basement exceeds 200 m^2 and **Note 9: S**prinklers shall be fed water from both underground static water storage tank and terrace tank)

45 m and above in height to installed in entire building. (Basements, ground and all upper floors)

		floor		
		1 st floor	52	--
		2 nd floor	44	--
		3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd& 25 th floor	45 on each floor (Total – 495 Nos.)	--
		19 th floor	43	--
		4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26 th floor	44 on each floor (Total – 528 Nos.)	
		Wing-F		
		Ground floor	42	--
		1 st floor	52	--
		2 nd floor	44	--
		3rd, 5th, 7th, 9th, 11th, 13th, 15th, 17th, 21st, 23rd& 25 th floor	44 on each floor (Total – 484 Nos.)	--
		19 th floor	43	
		4th, 6th, 8th, 10th, 12th, 14th, 16th, 18th, 20th, 22nd, 24th& 26 th floor	44 on each floor (Total – 528 Nos.	--
	NBC-2016, Part-4, Annex-E-4 HORIZONTAL EXITS/REFUGE AREA a) A horizontal exit shall be through a fire door of 120 min rating in a fire resistant wall. Horizontal exit require	**Building-1** **Wing-A & B** **Proposed provide Refuge area of 29.00 Sq. mtrs. at 19 th floor of Wing-A & 39.00 Sq. mtrs. at 19 th floor of Wing-B (Total – 68.00 Sq. mtrs.)** **As per Table 3 Occupant Load (Clauses 4.3 and 4.4.2.1), for Residential Building occupant load factor (Sq. mtrs/person) is 12.50. The calculation as follows:-**		

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14	separation with the refuge area or adjoining compartment through 120 min fire barrier. The adjoining compartment of the horizontal exit should allow unlocked and ease of egress and exits for the occupants using defend in place strategy. Requirements of horizontal exits are as under: a) Width of horizontal exit doorway shall be suitable to meet the occupant load factor for egress. b) Doors in horizontal exits shall be openable at all times from both sides. c) All doors shall swing in the direction of exit travel. For horizontal exits, if a double leaf door is used, the right hand door leaf shall swing in the direction of exit travel. 1) The refuge area shall be provided on the periphery of the floor and open to air at least on one side protected with suitable railings. 2) A prominent sign bearing the words 'REFUGE AREA' shall be installed at the entry of the refuge area, having height of letters of minimum 75 mm and also containing information about the location of refuge areas on the floors above and below this floor. The same signage shall also be conspicuously located within the refuge area. f) Each refuge area shall be ventilated and provided with first aid box, fire extinguishers, public address speaker, fire man talk back, and adequate emergency lighting as well as drinking water facility. g) Refuge areas shall be approachable from the space they serve by an accessible means of egress. h) Refuge areas shall connect to fire fighting shaft (comprising fireman.s lift, lobby and staircase) without having the occupants requiring to return to the building spaces through which travel to the area of refuge occurred. j) The refuge area shall always be kept clear. No storage of combustible products and materials, electrical and mechanical equipment, etc shall be allowed in such areas. k) Refuge area shall be provided with adequate drainage facility to maintain efficient storm water disposal. m) Entire refuge area shall be provided with sprinklers. n) Where there is a difference in level between connected areas for horizontal exits, ramps of slope not steeper than 1 in 12 shall be provided (and steps should be avoided). ***NOTE***. Refuge area provided in excess of the requirements shall be counted towards FAR. High rise apartment buildings with apartments having balcony, need not be provided with refuge area; however apartment buildings without balcony shall provide refuge area as given above. Refuge areas for apartment buildings of height	**Refuge area at 20 th floor at 60 mtrs. height** . 19 th floor Built-up area of Wing-A & B is 1,321.00 Sq. mtrs. / 12.5 = 105.68 persons. 20thfloor Built-up area of Wing-A & B is 1,295.70 Sq. mtrs. / 12.5 = 103.65 persons. **Total 209.33 persons x 0.3 Sq.mtrs. = 62.80 Sq.mtrs. + 1.80 Sq. mtrs. = 64.59 Sq. mtrs.** **Proposed Refuge area is more than the requirement.** **Building-2** **Wing-C & D** **Proposed provide Refuge area of 29.00 Sq. mtrs. at 19 th floor of Wing-A & 39.00 Sq. mtrs. at 19 th floor of Wing-B (Total – 68.00 Sq. mtrs.)** **As per Table 3 Occupant Load (Clauses 4.3 and 4.4.2.1), for Residential Building occupant load factor (Sq. mtrs/person) is 12.50. The calculation as follows:-** **Refuge area at 20 th floor at 60 mtrs. height** . 19 th floor Built-up area of Wing-A & B is 1,321.00 Sq. mtrs. / 12.5 = 105.68 persons. 20thfloor Built-up area of Wing-A & B is 1,295.70 Sq. mtrs. / 12.5 = 103.65 persons. **Total 209.33 persons x 0.3 Sq.mtrs. = 62.80 Sq.mtrs. + 1.80 Sq. mtrs. = 64.59 Sq. mtrs.** **Proposed Refuge area is more than the requirement.** **Building-3** **Wing-E & F** **Proposed provide Refuge area of 45.00 Sq. mtrs. at 19 th floor of Wing-E & 45.00 Sq. mtrs. at 19 th floor of Wing-F (Total – 90.00 Sq. mtrs.)** **As per Table 3 Occupant Load (Clauses 4.3 and 4.4.2.1), for Residential Building occupant**

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	above 60 m while having balconies shall be provided at 60 m and thereafter at every 30 m. The refuge area shall be an area equivalent to 0.3 m^2 per person for accommodating occupants of two consecutive floors, where occupant load shall be derived on basis of 12.5 m^2 of gross floor area and additionally 0.9 m^2 for accommodating wheel chair requirement or shall be 15 m^2, whichever is higher.	**load factor (Sq. mtrs/person) is 12.50. The calculation as follows:-** **Refuge area at 20 th floor at 60 mtrs. height .** 19 th floor Built-up area of Wing-E & F is 1,509.44 Sq. mtrs. / 12.5 = 120.75 persons. 20thfloor Built-up area of Wing-E & F is 1,462.36 Sq. mtrs. / 12.5 = 116.98 persons. **Total 237.73 persons x 0.3 Sq. mtrs. = 71.32 Sq.mtrs. + 1.80 Sq. mtrs. = 73.12 Sq. mtrs.** **Proposed Refuge area is more than the requirement.**
15	Fire Command Centre ***NBC-2016, Part-4 Clause 3.4.12 Fire Command Centre (FCC)*** a) Fire command centre shall be on the entrance floor of the building having direct access. The control room shall have the main fire alarm panel with communication system (suitable public address system) to aid floors and facilities for receiving the message from different floors. b) Fire command centre shall be constructed with 120 min rating walls with a fire door and shall be provided with emergency lighting. Interior finishes shall not use any flammable materials. All controls and monitoring of fire alarm systems, pressurization systems, smoke management systems shall happen from this room. Monitoring of integrated building management systems, CCTVs or any other critical parameters in building may also be from the same room. c) Details of all floor plans along with the details of fire fighting equipment and installations (2 sets laminated and bound) shall be maintained in fire command centre. d) The fire staff in charge of the fire command centre shall be responsible for the maintenance of the various services and fire fighting equipment and installations in coordination with security, electrical and civil staff of the building.	Proposed to provide Fire command centre at ground floor of each Wing as per NBC 2016.
	NBC-2016, Part-4, Annex-D, Clause 4.11 D-5 FIRE SAFETY PLAN ***D-5.1 A format for the Fire Safety Plan shall be as given in D-9.10.*** ***D-5.2*** The applicable parts of the approved Fire Safety Plan shall be distributed to all tenants of the building by the building management when the Fire Safety Plan has been approved by the Fire Authority.	

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16	D-5.3 The applicable parts of the approved Fire Safety Plan shall then be distributed by the tenants to all their employees and by the building management to all their building employees. D-5.4 In the event there are changes from conditions existing at the time the Fire Safety Plan for the building was approved, and the changes are such so as to require amending the Fire Safety Plan, within 30 days after such changes, an amended Fire Safety Plan shall be submitted to the fire brigade for approval.	FIRE SAFETY PLAN should be provided as per NBC 2016.
17	Fire Officer ***As per clause 4.10 of Part 4 Fire and Life Safety of NBC 2016:*** **4.10 Fire Officer** **4.10.1** A qualified Fire Officer with experience of not less than 3 years shall be appointed who will be available on residential building with height 60 m and above. **4.10.2** The Fire Officer shall, a)maintain the fire fighting equipment in good working condition at all times. b)prepare fire orders and fire operational plans and get them promulgated. c)impart regular training to the occupants of the buildings in the use of fire fighting equipment provided on the premises and keep them informed about the fire emergency evacuation plan. d)keep proper liaison with the city fire brigade. e)ensure that all fire precautionary measures are observed at the times. NOTE - Competent authority having jurisdiction may insist on compliance of the above rules in case of buildings having very large areas even if the height is less than 30 m.	Fire Officer should be appointed as per Part-4 of NBC 2016. Training - 40% of the occupants/employees of each block should be got trained in fire prevention & fire fighting at the R.A. Mundkur Fire and Emergency Services Academy, Bannerghatta Road, Bangalore-560 029, within 6 months from the date of occupation of the Building. For this purpose before approaching this department for final clearance certificate, the applicant should give an undertaking in the form of an affidavit regarding the maintenance of the fire prevention and fire fighting measures suggested above and arranging training of 40% of the occupants in fire prevention and fire fighting within 6 months from the date of issue of the Clearance Certificate.
		As proposed Fire extinguishers at following suitable places should be provided. 1) One ABC powder extinguishers of 6 kgs. and9Litres capacity Portable Hand held "Water Mist & CAFs" fire extinguishers – Jet & spray (combination) Capacity for every 8 cars at parking areas should be provided. - As per IS-15683 / EN3-7 / NFPA-10 (Design & Construction) - Suppression Technology: NFPA 750 & NFPA 11 - Minimum Lancing Distance : Jet – 30Feet or more.

	- Minimum Lancing Distance : Spray– 10Feet or more - Spraying Angle 60° - Class – A, B, LPG Fires and Live Electrical Fire below 1000Volts (Test certificate to be submitted) - Fire Rating A: 21A or more - Fire Rating B: 144B or more - Foam Mist / Pressurised bubbles (Adhere to vertical and Horizontal surfaces, bubbles should retain for a minimum period of 20Minutes) - Foam Expansion minimum 1:10 or more **2)**One CO_2 extinguishers of 4.5 kgs. Capacity should be provided near the entrance to the electrical room. **3)**One 9 Litres capacity Portable Hand held "Water Mist & CAFs" fire extinguishers – Jet / spray type (Combination) and One ABC powder extinguishers of 6kgs. Capacity should be provided near transformer. - As per IS-15683 / EN3-7 / NFPA-10 (Design & Construction) - Suppression Technology: NFPA 750 & NFPA 11 - Minimum Lancing Distance : Jet – 30Feet or more - Minimum Lancing Distance : Spray– 10Feet or more - Spraying Angle 60° - Class – A, B, LPG Fires and Live Electrical Fire below 1000Volts (Test certificate to be submitted) - Fire Rating A: 21A or more - Fire Rating B: 144B or more - Foam Mist / Pressurised bubbles (Adhere to vertical and Horizontal surfaces, bubbles should retain for a minimum period of 20Minutes) - Foam Expansion minimum 1:10 or more and one ABC powder extinguishers of 6 kgs. Capacity should be provided near the diesel generator.
Fire extinguishers ***NBC-2016, Part-4, Table 7 (3) Fire extinguishers shall be provided for,*** ***Apartment Houses (A-4)*** 1. One ABC powder extinguishers of 6 kgs. Capacity for every 8 cars at parking areas should be provided. 2. One CO_2 extinguishers of 4.5 kgs. Capacity should be provided near the entrance to the electrical room. 3. One Mechanical Foam extinguishers of 9 litres capacity & one ABC powder extinguishers of 6 kgs. Capacity should be provided near the transformer. 4. One Mechanical foam extinguishers of 9 litres	

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18	4. One Mechanical foam extinguishers of 9 litres capacity and one ABC powder extinguishers of 6 kgs. Capacity should be provided near the diesel generator. 5. One CO_2 extinguishers of 2 kgs. Capacity should be provided inside each lift machine room. 6. One CO_2 extinguishers of 2 kgs. Capacity should be provided inside each kitchen. 7. One Water Mist type extinguishers of 4 litres & 9 litres capacity should be kept near each staircase landing at each floor. All the extinguishers suggested above should be with B.I.S. markings and should be located at an easily accessible position without obstructing the normal passage and maintained periodically.	**4)**One 9Litres capacity Portable Hand held "Water Mist & CAFs" fire extinguishers – Jet / spray type (Combination) and One ABC powder extinguishers of 6kgs. Capacity should be provided near the diesel generator. - As per IS-15683 / EN3-7 / NFPA-10 (Design & Construction) - Suppression Technology: NFPA 750 & NFPA 11 - Minimum Lancing Distance : Jet – 30Feet or more - Minimum Lancing Distance : Spray– 10Feet or more - Spraying Angle 60° - Class – A, B, LPG Fires and Live Electrical Fire below 1000Volts (Test certificate to be submitted) - Fire Rating A: 21A or more - Fire Rating B: 144B or more - Foam Mist / Pressurised bubbles (Adhere to vertical and Horizontal surfaces, bubbles should retain for a minimum period of 20Minutes) - Foam Expansion minimum 1:10 or more and one ABC powder extinguishers of 6 kgs. Capacity should be provided near the diesel generator. **5)**One CO_2 extinguishers of 2 Kgs. Capacity should be provided inside each lift machine room. **6)**9Litres capacity Portable Hand held "Water Mist & CAFs" fire extinguishers – Jet / spray type (combination) should be kept in alternative staircase landing at each floor. - As per IS-15683 / EN3-7 / NFPA-10 (Design & Construction) - Suppression Technology: NFPA 750 & NFPA 11 - Minimum Lancing Distance : Jet – 30Feet or more - Minimum Lancing Distance : Spray– 10Feet or more - Spraying Angle 60° - Class – A, B, LPG Fires and Live

		Electrical Fire below 1000Volts (Test certificate to be submitted) - Fire Rating A: 21A or more - Fire Rating B: 144B or more - Foam Mist / Pressurised bubbles (Adhere to vertical and Horizontal surfaces, bubbles should retain for a minimum period of 20Minutes) - Foam Expansion minimum 1:10 or more All the extinguishers suggested above should be with B.I.S. markings and should be located at an easily accessible position without obstructing the normal passage and maintained periodically.
19	As per clause 3.4.3 of Part-4 Fire and Life Safety of NBC-2016: Occupation of Buildings under Construction 3.4.3.2 A high rise building during construction shall be provided with the following fire protection measures, which shall be maintained in good working condition at all the times: a) Dry riser of minimum 100 mm diameter pipe with hydrant outlets on the floors constructed with a fire service inlet to boost the water in the dry riser and maintenance should be accordance with good practice. b) Drums of 2000 litre capacity filled with water with two fire buckets on each floor. c) A water storage tank of minimum 20000 litre capacity, which may be used for other construction purposes also	As per clause 3.4.3 of Part-4 Fire and Life Safety of NBC-2016, the below mentioned fire precautionary measures should be taken during the time of construction of this building and should be maintained in good working condition in all the times. a) Dry riser of minimum 100 mm diameter pipe with hydrant outlets on the floors constructed with a fire service inlet to boost the water in the dry riser and maintenance should be in accordance with good practice [4(5)]. b) Drums of 2000 litres capacity filled with water with two fire buckets on each floor. c) Water storage tank of minimum 20,000 litres capacity, which may be used for other construction.

CONDITIONS:-

1. All the fire prevention, fire fighting and evacuation measures suggested/ recommended in B, C and D shall be strictly adhered to adopted.
2. Hazardous materials such as petroleum products, explosives, chemicals etc. should not be stored on any floor.
3. Refuse dumps or storage should not be permitted in any of the floors.
4. Liquefied petroleum gas should not be stored in the building except the limited quantity required for each kitchen.
5. Plan and occupancy should not be changed without informing the Fire & Emergency Services and without taking clearance.
6. The occupancy certificates should not be issued without obtaining the clearance certificate from the Fire and Emergency Services department.

7. Such reasonable changes/modifications as may be found necessary, after the building is fully constructed, will have to be agreed to be done by the builder/occupants of the building.
8. All the metal fittings of down comer system and all the extinguishers suggested above should have B.I.S markings.
9. Apart from the above the Building shall be constructed by following all the rules & conditions stipulated in Part-III & IV of NBC & local zoning regulations strictly, failing which the NOC issued will not be valid.
10. The above mentioned requirements are indicative and not exhaustive. All other requirements of National Building Code not specifically mentioned above shall also be complied with mandatorily.
11. This NOC is issued from the Fire Prevention and Fire Fighting point of view Karnataka State Fire & Emergency Services Department is not responsible for the ownership of the land, its location and other disputes, which may arise in due course.

Subject to the strict adherence to the conditions laid down as above, issue of License for the construction Residential that is AT SY NO. 55/1, 55/2, 55/3, 55/4, 56/2, 56/3, 62/1, 62/4, 62/5, 63/2, 63/3, 63/4, 63/5, 63/6 GUNJUR VILLAGE, VARTHUR HOBLI, Bangalore East, BANGALORE - 560087 may please be considered.

- **All other relevant and applicable requirements as per NBC-2016 will have to be compiled with mandatorily.**

Yours faithfully,
Director General of Police and Director,
Karnataka Fire & Emergency Services.

Digitally signed by T N SHIVASHANKARA
Date: 2025.03.07 11:36:34 +05:30
Reason: Approved